WORKED SOLUTION

STEP II 2004, Question 11

Mechanics • Work and power

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Read the question before you go on. It is in the STEP Support Programme's question database:

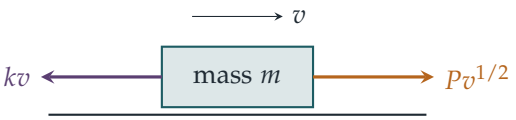
step.maths.org/questions/04-s2-q11

The key idea

Power is force times speed, so an engine working at full power $Pv^{3/2}$ pushes with a force $Pv^{1/2}$. Newton's second law then gives a differential equation with a $\sqrt{v}$ in it, and the substitution $u = \sqrt{v}$ turns it into a linear first-order equation that we can solve at once. After the engines are switched off, each speed decays exponentially, and the trick in part (iii) is to compare the two decay rates.

Solution

We model each train as a particle moving along a straight horizontal track. While the engine is on at full power, the forces along the track are the driving force and the resistance kv.



If an engine develops power H at speed v, its driving force is H/v. So at full power the driving force on A is $Pv^{3/2}/v = Pv^{1/2}$, and on B it is $2Pv^{1/2}$.

Part (i)

For A, Newton's second law along the track gives

$$m\frac{\mathrm{d}v_A}{\mathrm{d}t} = Pv_A^{1/2} - kv_A.$$

Put $u = v_A^{1/2}$, so that $v_A = u^2$ and $\dfrac{\mathrm{d}v_A}{\mathrm{d}t} = 2u\dfrac{\mathrm{d}u}{\mathrm{d}t}$. Then

$$2mu\frac{\mathrm{d}u}{\mathrm{d}t} = Pu - ku^2, \qquad \text{so} \qquad 2m\frac{\mathrm{d}u}{\mathrm{d}t} = P - ku,$$

dividing by u, which is allowed because the train is always moving ($u > 0$). This is linear (and separable):

$$\int \frac{2m\,\mathrm{d}u}{P-ku} = \int \mathrm{d}t \quad \Longrightarrow \quad -\frac{2m}{k}\ln(P-ku) = t + \text{const},$$

so $P - ku = Ce^{-kt/2m}$ for some constant C. (Here $P - ku > 0$ at the start, when u is tiny, and it stays positive because the solution never reaches the equilibrium $u = P/k$.)

Ask yourself What does "crawling along at very low speeds" tell us?

It gives the initial condition. The speeds at $t = 0$ are positive (so dividing by u was fine) but negligibly small, so we take $u = 0$ at $t = 0$. Then $C = P$, and

$$u = \frac{P}{k}\left(1 - e^{-kt/2m}\right), \qquad \text{so} \qquad v_A = \frac{P^2\left(1 - e^{-kt/2m}\right)^2}{k^2},$$

as required. (If the train were exactly at rest, $v \equiv 0$ would also satisfy the original equation, since there would be no driving force at all; this is why the question has the trains already moving slowly.)

For B the equation is $2m\dfrac{\mathrm{d}v_B}{\mathrm{d}t} = 2Pv_B^{1/2} - kv_B$. The same substitution gives $4m\dfrac{\mathrm{d}u}{\mathrm{d}t} = 2P - ku$, which is the equation for A with P replaced by $2P$ and m by $2m$. So

$$v_B = \frac{4P^2\left(1 - e^{-kt/4m}\right)^2}{k^2}.$$

Part (ii)

Both speeds are positive, so $9v_A = 4v_B$ is equivalent to $3\sqrt{v_A} = 2\sqrt{v_B}$. Write $x = e^{-kt/4m}$, so that $e^{-kt/2m} = x^2$, and $0 < x < 1$ for $t > 0$. The condition becomes

$$\frac{3P}{k}\left(1 - x^2\right) = \frac{4P}{k}(1 - x) \quad \Longrightarrow \quad 3(1-x)(1+x) = 4(1-x).$$

Since $x \neq 1$ we may divide by $1 - x$, giving $3 + 3x = 4$, so $x = \frac{1}{3}$ (that is, $t = \frac{4m}{k}\ln 3$). Then $1 - x^2 = \frac{8}{9}$ and $1 - x = \frac{2}{3}$, so

$$v_A = \frac{64P^2}{81k^2}, \qquad v_B = \frac{16P^2}{9k^2}.$$

As a quick check, $9v_A = \frac{64P^2}{9k^2} = 4v_B$.

Part (iii)

Measure time τ from the moment the engines are switched off, and let $V_A = \frac{64P^2}{81k^2}$ and $V_B = \frac{16P^2}{9k^2}$ be the speeds at that moment. Now only resistance acts:

$$m\frac{\mathrm{d}v_A}{\mathrm{d}\tau} = -kv_A, \qquad 2m\frac{\mathrm{d}v_B}{\mathrm{d}\tau} = -kv_B,$$

so

$$v_A = V_A\,\mathrm{e}^{-k\tau/m}, \qquad v_B = V_B\,\mathrm{e}^{-k\tau/2m}.$$

The heavier train B loses speed at half the exponential rate, so squaring v_B matches the exponentials:

$$v_B^2 = V_B^2\,\mathrm{e}^{-k\tau/m} = \frac{V_B^2}{V_A}\,v_A.$$

Finally

$$\frac{V_B^2}{V_A} = \frac{256P^4}{81k^4}\cdot\frac{81k^2}{64P^2} = \frac{4P^2}{k^2},$$

so $v_B^2 = \dfrac{4P^2}{k^2}v_A$, that is, $k^2v_B^2 = 4P^2v_A$ for all later times, as required.

Checks

- As $t\to\infty$, $v_A\to P^2/k^2$, which is where the driving force $Pv^{1/2}$ equals the resistance kv. Similarly $v_B\to 4P^2/k^2$, where $2Pv^{1/2}=kv$.
- For small t, $v_A\approx P^2t^2/(4m^2)$ and $v_B\approx P^2t^2/(4m^2)$: the trains start off together, as they should, since each has driving force per unit mass $Pv^{1/2}/m$.
- With $m=1.7$, $P=2.3$, $k=0.9$ we stepped both equations of motion numerically (Runge–Kutta) from a tiny initial speed and matched the formulae of part (i) to about 10^{-5}; the moment $9v_A=4v_B$ gave $\mathrm{e}^{-kt/4m}=0.33333$, $v_A=5.1602$ and $v_B=11.6104$, and stepping on with the engines off kept $k^2v_B^2$ equal to $4P^2v_A$ (for example both 37.8739 two seconds later).

Common slips

- Using $Pv^{3/2}$ as the force. It is a power; the force is the power divided by the speed.
- Forgetting that the condition in part (ii) is a quadratic in $x=\mathrm{e}^{-kt/4m}$ with the root $x=1$ (that is, $t=0$, when both speeds are zero) as well as $x=\frac{1}{3}$.

Beyond STEP

The substitution $u=\sqrt{v}$ is a special case of the method for Bernoulli equations, $y'+p(t)y=q(t)y^n$, where $u=y^{1-n}$ always gives a linear equation. The remark about the train exactly at rest is a first glimpse of uniqueness theory: because $\sqrt{v}$ has infinite slope at $v=0$, the equation has more than one solution from $v=0$, which the Picard–Lindelöf theorem explains. Boyce and DiPrima, *Elementary Differential Equations*, covers both; for the mechanics of power and resistance, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

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